Problem Statement

A **leading retail company** wants to better **understand** its **customers' shopping behavior** in order to **improve sales, customer satisfaction, and long term loyalty**. The **management team** has noticed **changes** in **purchasing patterns** across **demographics(like age, gender,etc), product categories, and sales channels** (online vs. offline). They are particularly **interested in uncovering** which **factors**, such as **discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.**

You are **tasked with analyzing** the **company's consumer behavior dataset** to **answer** the following overarching **business question**:

"**How** can the **company leverage consumer shopping data** to **identify trends, improve customer engagement, and optimize marketing and product strategies?"**

Deliverables:

1. **Data Preparation & Modeling (Python): Clean** and **transform** the raw **dataset** for **analysis**.
2. **Data Analysis (SQL): Organize** the **data into** a **structured format**, **simulate business transactions**, and **run queries** to **extract insights** on **customer segments, loyalty, and purchase drivers.**
3. **Visualization & Insights (Power BI): Build** an interactive **dashboard** that **highlights key patterns and trends**, **enabling stakeholders** to make **data-driven decisions**.
4. **Report and Presentation: Write** a clear **project report summarizing** your **key findings and business recommendations**. Prepare a **presentation** that **visually communicates insights** and **actionable recommendations** to stakeholders.
5. **GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

Business questions:

```
SELECT * FROM customer_behavior.customer;

describe customer;

ALTER TABLE customer

RENAME COLUMN `purchase_amount_(usd)` TO purchase_amount; --  backticks i.e. ` when special character present in col name


-- Q1. What is the total revenue generated by male vs. female customers?

select gender, sum(purchase_amount) as revenue

from customer

group by gender;
```

```
-- 'Q2. Which customers used a discount but still spent more than the average purchase amount? '
select customer_id, purchase_amount
from customer
where purchase_amount > (select avg(purchase_amount) from customer)
and discount_applied='yes';

-- 'Q3. Which are the top 5 products with the highest average review rating?'
select item_purchased, round(avg(review_rating),2) as Average_product_rating
from customer
group by item_purchased
order by Average_product_rating desc
limit 5;

-- 'Q4. Compare the average Purchase Amounts between Standard and Express Shipping. '
select shipping_type, round(avg(purchase_amount),2) as avg_purchase_amount
from customer
where shipping_type in ('standard', 'express')
group by shipping_type;
#------------- here, avg_purchase_amount is higher for express than standard so, business should
#---------------focus on express shiping and other faster delivery options.

-- 'Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.'
select subscription_status, count(customer_id) as total_customers,
round(avg(purchase_amount),2) as avg_spend, round(sum(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by total_revenue, avg_spend desc;

-- 'Q6. Which 5 products have the highest percentage of purchases with discounts applied?'
select item_purchased, round(100 * sum(case when discount_applied= 'yes' then 1 else 0 end)/count(*),2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

-- 'Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
select (CASE
        WHEN previous_purchases = 1 THEN 'New'
   WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
   ELSE 'Loyal'
   END) AS customer_segment, count(*) as 'number of customers'
from customer
group by customer_segment;
```

-- .....solve using cte

```
with customer_type as (
select customer_id, previous_purchases,
CASE
        WHEN previous_purchases = 1 THEN 'New'
   WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
   ELSE 'Loyal'
   END AS customer_segment
from customer)

select customer_segment, count(*) as 'number of customers'
from customer_type
group by customer_segment;
```

-- 'Q8. What are the top 3 most purchased products within each category? '

-- .....required window functions and cte

```
with item_counts as (select category, item_purchased, count(customer_id) as total_orders,
ROW_NUMBER() over(partition by category order by count(customer_id) desc) as item_rank
from customer
group by category, item_purchased)

select item_rank, category, item_purchased, total_orders
from item_counts
where item_rank <= 3;
```

```
-- ..............................................Without CTE
SELECT *
FROM (
SELECT category,
    item_purchased,
    COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER(
        PARTITION BY category
        ORDER BY COUNT(customer_id) DESC
    ) AS rank_num
FROM customer
GROUP BY category, item_purchased
) t
WHERE rank_num <= 3;

-- 'Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?'
select subscription_status, count(customer_id) as repeat_buyers
from customer
where previous_purchases>5
group by subscription_status;

-- 'Q10. What is the revenue contribution of each age group?'
select age_group, sum(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc;
```